

Jonas Immanuel Frey

AI & Full-Stack Software Engineer

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Applying for: AI Software Engineer — Avaloq AI Lab

Profile

Relentless, self-driven engineer who turns ideas into shipping software — fast. Across **199 public repositories** I have built and released full-stack web apps, real AI/ML inference pipelines, developer tooling with real users, and even hardware controllers, spanning **JavaScript/TypeScript, Deno, Python, Rust and C#**. I learn new stacks by building production-grade things with them, not by waiting for permission.

What I bring to the Avaloq AI Lab is exactly the "ideation to production" mindset the role asks for: I have personally taken concepts from a blank folder to a deployed, running product — including **AI systems that transcribe, analyse and act on real-world data**. I don't yet hold a Master's or PhD, but I have an unusually deep, hands-on track record of *building and shipping*, and I move quickly from "I've never used this" to "I shipped it."

Core Skills

Area	Technologies & Evidence
Languages	Python · JavaScript · TypeScript · Rust · C# · Shell · G-code/OpenSCAD
AI / ML	OpenAI Whisper (speech-to-text inference), PyTorch , pose estimation & computer vision, AI image alignment/stacking, prompt engineering, GPU compute (WebGL/GPU shaders)
Backend & APIs	Self-built HTTP + WebSocket micro-service framework (published package), REST APIs, real-time WebSocket services, SQLite
Frontend / Full-Stack	Vue 3, isomorphic native-JS app framework, browser tooling & extensions
DevOps / Infra	Cloud deployment (Deno Deploy), GPU observability tooling (nvidia-smi / amdgpu), Linux, Git
Hardware / Systems	Raspberry Pi (Rust), ESP32/IoT, stepper-motor & robotics control, desktop automation

Selected Projects

(mapped to what the AI Lab actually does — AI from ideation to production, in a full-stack context)

🎬 Video Wizard — AI-driven media analysis pipeline

github.com/jonasfrey/video_wizard An end-to-end AI application: video → audio extraction → **speech recognition (OpenAI Whisper / PyTorch)** → text-based editing → automated render/export. Full-stack: **Deno** server, **Vue 3** GUI, **WebSocket** streaming, **FFmpeg** and **Python** inference scripts. *Demonstrates: shipping an AI inference product from ideation to a running application — model integration, Python, full-stack, orchestration of moving parts.*

📊 Browser GPU Monitor — ML-infra observability tool (★ 21)

github.com/jonasfrey/gpu-monitor-browser-gui A browser dashboard for live GPU telemetry (NVIDIA `nvidia-smi` / AMD `amdgpu_top`), used by real people (21 stars). *Demonstrates: observability of AI compute — directly relevant to LLM/Agent Ops, monitoring and evaluation infrastructure.*

🔗 websocket — micro-service & REST framework

github.com/jonasfrey/websocket · published to the JSR registry as `@apn/websocketgui` A reusable Deno framework for spinning up HTTP file servers, **REST endpoints**, **reverse proxies** and **WebSocket services** — the backbone I reuse across many apps (`iot_garden` , `tee_dunker` , and more). *Demonstrates: micro-services, REST APIs, real-time services, and packaging/publishing reusable infrastructure.*

🤖 AI vision & automation

[ai_stacking](#) (AI image alignment & stacking) · [pilotless_poses2](#) (pose estimation) · [robotjs_proxy](#) (desktop automation). *Demonstrates: applied computer vision and agent-style automation of real actions.*

[ruspi_microscope](#) — Rust + Raspberry Pi control

[github.com/jonasfrey/ruspi_microscope](#) Low-level control of a motorised microscope stage in **Rust** on Raspberry Pi. *Demonstrates: systems programming, precision, and comfort across the full stack down to the metal.*

Also relevant: [prompt_engineer_ds](#) (prompt engineering), [machine_learning](#), [CADprogramming](#) & [blender_auto_render](#) (Python), [cybersecurity_lessons](#) / [cybersecurity_project](#) (security awareness — important in banking), plus dozens of shipped web tools deployed to the cloud.

Experience

Independent Software Engineer & Open-Source Builder — *ongoing* Design, build, ship and maintain a large portfolio of applications end-to-end: AI/ML pipelines, full-stack web platforms, developer tooling and hardware controllers. Full ownership from ideation → architecture → implementation → deployment → maintenance.

Professional roles & exact dates: to be completed by Jonas.

Education

- **Bern University of Applied Sciences (BFH)** — Computer Science studies (*please confirm degree & years — see [bfh_java_kotlin](#)*)
- **GIBB Berufsfachschule Bern** — Vocational IT training (Informatiker EFZ) (*please confirm specialisation & years — see [gibb_hf_abschlusspruefung_applikation](#), [GIBB_DEP_modul](#), [GIBB_WEB1_projekt](#)*)
- **Continuous self-education** — 199 shipped repositories as a living, public learning record

Note on the "B.Sc." requirement: my strongest qualification is a large, verifiable body of production work. If a completed degree is essential, I'm happy to discuss my current standing and timeline.

Languages

German — native · **English** — professional working proficiency

Why Avaloq AI Lab

I'm excited by the Lab's mission to bring **agentic AI from ideation into production** for real banking users. That is precisely the loop I already run solo — build, deploy, iterate — and I'm eager to do it inside a focused team, at a Swiss industry leader, with the scale and rigour that banking demands. I'm a fast learner with a proven record of picking up new stacks (agent frameworks, RAG, vector stores, cloud ML services) and shipping with them.